BVCC Wallet

BVCC Agent Wallet

A non-custodial wallet that gives an AI agent spending limits the contract itself enforces.

Open source · ERC-4337 + passkeys · Five mainnets · Built by BlockVenture Chain Capital

PUBLIC BETA

NON-CUSTODIAL · GPL-3.0

NO KYC · SELF-HOSTABLE

THE PROBLEM

To let an AI agent transact today, you give it a private key. A private key has no ceiling. If the agent is prompt-injected, hallucinates a destination, or the key leaks out of a config file, everything in that wallet is gone, and nothing on-chain was in a position to stop it.

The usual answer is a server in the middle that decides which transactions to sign. That works, and it moves the problem. Now you trust the server, and whoever runs it can sign whatever it likes.

BVCC puts the limits in the contract instead. A leaked agent key is worth its remaining budget, not the account.

WHAT IS ACTUALLY DIFFERENT

Most work on agent wallets stops at a spending cap. The harder problem is that a cap says nothing about intent. An agent authorized to swap on Uniswap with a $500 budget can route that swap so the output lands in an attacker's address, and it never goes over $500.

So the wallet measures what actually leaves. Before and after every agent operation it reads the balance of each token involved, and whatever left is charged to the agent's budget, fee included, whatever route the agent picked. Every DeFi call also passes an on-chain validator: for a Universal Router swap, the command bytes must match exactly and every recipient must be your wallet. The agent can trade. It cannot trade on someone else's behalf, or spend past its budget through an old approval.

WHAT AN AGENT CAN ACTUALLY DO

Lending — Aave v3 — 19 tools

Supply, borrow, repay, withdraw. Plus four planners that unwind a live position: deleverage, close, swap the collateral, swap the debt — each respecting a health-factor floor and aborting on oracle deviation.

Liquidity — Uniswap v3 & v4 — 14 tools

Open a position at a chosen range, collect fees, reduce, close. v4 native ETH pools included, with hooks gated by a registry on a 48-hour timelock.

Swaps & transfers — 22 tools

Uniswap v3 and v4, native ETH in and out, ERC-20 and native sends, approvals.

Before it acts — 17 simulate

Most writes have a matching dry-run returning gas and the revert reason, so a model can check whether something works before spending. Four guides it can read first.

TRY IT WITHOUT RISKING ANYTHING

```
$ BVCC_MCP_READONLY=true npx -y @bvcc/agent-mcp
▸ 29 tools registered · all writes removed
▸ read balances · quote swaps · simulate — cannot move funds
```

Drop the flag and the same server exposes all 57 tools. Any MCP client works: Claude, Cursor, LM Studio, Hermes. Every call still has to pass the contract.

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MEDIA KIT · REFERENCE

BY THE NUMBERS

191 Foundry tests on V5 503 counting the V4 suite	**57** MCP tools 12 read · 17 sim · 28 write	**29** tools in read-only mode every write removed
6 networks, one address deterministic via CREATE2	**72** V5 contracts all source-verified	**0** databases, emails or KYC the passkey derives the address

SECURITY, IN FULL

Critical	2	both fixed
High	13	12 fixed · 1 accepted
Medium	10	8 fixed · 2 accepted
Low	3	2 fixed · 1 accepted
Informational	2	both fixed

Internal review of every generation; the V5 rounds were run with AI tools. **Not externally audited.** 30 findings, status on V5: nothing open, four accepted as stated trust assumptions. V5 closes BVCC-03, still open on V4 wallets. Full 61-page report is published.

LIMITS THE CONTRACT ENFORCES

- **Budget** per period with auto-rollover, plus daily, per-transaction and lifetime caps
- **Per-token** daily and total budgets, kept by address across re-authorization
- **Allowlists** for tokens, protocols and recipients; 0 or empty allows nothing; agents must be EOAs
- **Measured outflow**: what leaves is charged, fee included; every DeFi call has a validator
- **Expiry within a year and pause**, per agent or all at once, signed with your passkey
- **Recovery** by 2-of-3 guardians on a 48-hour timelock; completing one pauses every agent

PROVEN ON MAINNET (V4 AGENT WALLET)

Swap	Agent swap on Arbitrum through the Universal Router. Fee came out at 0.15% to the wei.
Lending	A full Aave v3 cycle by the agent: supply → borrow → repay → withdraw. Position closed.
Liquidity	Uniswap v4 ETH/USDC native pool: mint → collect → decrease → burn.
Blocked	A swap redirected to another address reverted with `PinnedArgMismatch()`.

STACK

Auth	WebAuthn / passkeys, P256
Account	ERC-4337 · ERC-7821 batching · ERC-7739 nested signatures
Contracts	Solidity 0.8.36, Foundry, OpenZeppelin. Frozen toolchain for deterministic CREATE2
App	Next.js 16, viem, wagmi. English and Spanish
Agents	Model Context Protocol · @bvcc/agent-sdk, @bvcc/agent-mcp (MIT)
Chains	Ethereum · Arbitrum One · Base · BNB Chain · Polygon · Arbitrum Sepolia

Site bvccwallet.blockventurechaincapital.com **Press** contact@blockventurechaincapital.com

Code github.com/blockventurechaincapital-crypto **Demo** youtu.be/dWUTaWBk68A

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